Cycle Sequencing

WHAT YOU GET

One primer, so extension is **linear**, not exponential. The read starts **20–50 bp downstream** of the primer and gives **400–1000 bp** of usable sequence. Pick a primer **upstream** of what you want to see.

For pP6 that is **G00101**.

USE THE DGTP PROTOCOL

Not the standard chemistry. Standard chemistry **dies inside a hairpin** — a terminator, or any strong secondary structure, stops the read dead. This cost Tlib2 an entire sequencing run.

Say **dGTP** on the submission form.

SUBMIT

1. Combine _____ µL miniprep DNA and _____ µL G00101, to _____ µL total.
2. Label each tube with its clone ID.
3. Specify the dGTP protocol.
4. Submit. Results come back in **1–2 days**.

> **These volumes are not written down anywhere.** Get them from your supervisor before you set the reactions up — then write them on this card, so the next person does not have to ask.

SEQUENCE EVERYTHING YOU PICKED

Including the boring-looking ones. The clones in the **middle of the range** are the informative ones. Tlib2 sequenced only its brightest and left its main question unanswered for three years.

CHECK THE READ

You get a `.txt` of base calls (the *read*) and an `.ab1` trace. Open both in **ApE**; `ctrl-K` annotates features.

1. Is it **clean**? Length with no Ns: 100 bp poor, 800 good, 1000 great.
2. Architecture **BseRI** → **promoter** → **BseRI**. Exactly **two** BseRI sites.
3. **T4 terminator** present; promoter not duplicated, reversed or truncated.
4. Align to `pP6.seq` — **Tools** → **Align with another sequence...** Look for 100% identity around the promoter.
5. Search for the motif below. Clean read + motif present = **usable**.

```
GAGGAGTCCTGGGTTCNNNNTTGACANNNNNNNNNNNNNNNNNTATAA
TNNNNNNANNNNGTTAGTATTTCTCCTC
```

RECORD

exp · **clone_id** · **student_name** · **read_name** · **date_sequenced** · **canonical** · **usable** · **cassette** · **notes**

- **canonical** = matches `pP6.seq` across the good-quality region. **usable** = contains the motif. Different questions; a clone can be one and not the other.
- Expect artifacts — the N-rich region throws up duplications, deletions and recombinations, and some reads come back as plain parent plasmid. Record them as artifacts and exclude them rather than forcing them in.